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1. PROBLEM STATEMENT

The rapid growth of e-commerce has resulted in the generation of a large amount of sales and order-related data. Analyzing this data efficiently can help identify sales patterns, order statuses, product categories and fulfilment-related information. In this project, the Amazon Sale Report dataset is analyzed using Apache Hadoop. The main objective is to demonstrate how Hadoop MapReduce and Apache Pig can be used to process and analyze e-commerce sales data.

The project focuses on performing different analyses, including category-wise total sales, status-wise order count, fulfilment-wise total quantity, category-wise total quantity, category-wise order count, category-wise average amount, top 10 categories based on quantity and category-wise maximum amount. This analysis provides a structured understanding of the sales data and demonstrates the practical use of Hadoop-based big data processing techniques.

2. DATASET DESCRIPTION

2.1 Dataset : <https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data>

The dataset used in this project is the Amazon Sale Report dataset, which contains e-commerce sales and order-related information. The dataset provides information about orders, products, quantities, sales amounts, order status, fulfilment, shipping details, promotions and customer-related order attributes. The dataset is stored in CSV format as Amazon Sale Report.csv. It contains 24 attributes and 128,975 data records, excluding the CSV header. These attributes provide the required information for performing different e-commerce sales analyses using Apache Hadoop MapReduce and Apache Pig. The dataset contains both categorical and numerical attributes. Among them, Qty represents the quantity of products ordered, while Amount represents the sales amount. Attributes such as Category, Status, Fulfilment, Sales Channel and Courier Status are useful for analyzing different aspects of the sales and order data.

2.2 Dataset Attributes

No.	Attribute	Description
1	index	Index number of the record
2	Order ID	Unique identifier of an order
3	Date	Date of the order
4	Status	Status of the order
5	Fulfilment	Fulfilment method used for the order
6	Sales Channel	Channel through which the order was placed
7	ship-service-level	Shipping service level

8	Style	Style identifier of the product
9	SKU	Stock Keeping Unit identifier
10	Category	Category of the product
11	Size	Size of the product
12	ASIN	Amazon Standard Identification Number
13	Courier Status	Status related to courier/shipping
14	Qty	Quantity of products ordered
15	currency	Currency used for the transaction
16	Amount	Sales amount of the order
17	ship-city	City where the order is shipped
18	ship-state	State where the order is shipped
19	ship-postal-code	Postal code of the shipping location
20	ship-country	Country where the order is shipped
21	promotion-ids	Promotion information associated with the order
22	B2B	Indicates whether the order is a B2B transaction
23	fulfilled-by	Entity responsible for fulfilling the order
24	Unnamed: 22	Additional/unnamed field present in the original dataset

3. SUMMARY OF DATA ANALYSIS

This project analyzes the Amazon Sale Report dataset using Apache Hadoop technologies, mainly MapReduce and Apache Pig. The analysis is designed to extract useful information from e-commerce sales and order data.

The MapReduce part of the project performs three major analyses:

- **Category-wise Total Sales** – calculates the total sales amount for each product category.
- **Status-wise Order Count** – counts the number of orders for each order status.
- **Fulfilment-wise Total Quantity** – calculates the total quantity of products handled by each fulfilment method.

The Apache Pig part performs five additional analyses:

- **Category-wise Total Quantity** – calculates the total quantity of products for each category.
- **Category-wise Order Count** – counts the number of orders for each category.
- **Category-wise Average Amount** – calculates the average sales amount for each category.
- **Top 10 Categories** – identifies the top 10 product categories based on total quantity.
- **Category-wise Maximum Amount** – finds the maximum sales amount for each product category.

These analyses provide different views of the e-commerce sales data and demonstrate how Hadoop MapReduce and Apache Pig can be used to process and analyze a large dataset efficiently.

4. MAPREDUCE ANALYSIS

Apache Hadoop MapReduce is a distributed data processing framework used to process large datasets by dividing the data processing task into two main phases: Map and Reduce. In the Map phase, input data is processed and converted into intermediate key-value pairs. In the Reduce phase, these intermediate values are grouped and aggregated to generate the final results.

In this project, Hadoop MapReduce is used to analyze the Amazon Sale Report dataset. Three different analyses are performed:

- ✓ **Category-wise Total Sales**
- ✓ **Status-wise Order Count**
- ✓ **Fulfilment-wise Total Quantity**

The dataset is stored in HDFS and separate Mapper, Reducer, and Driver programs are used for each analysis.

4.1 Execution

```
C:\Windows\System32>cd \  
  
C:\>cd hadoop  
  
C:\hadoop>cd sbin  
  
C:\hadoop\sbin>start-dfs.cmd  
  
C:\hadoop\sbin>start-yarn.cmd  
starting yarn daemons  
  
C:\hadoop\sbin>jps  
16592 Jps  
17616 DataNode  
3728 NodeManager  
8692 ResourceManager  
1128 NameNode  
  
C:\hadoop\sbin>hdfs dfs -mkdir /project  
  
C:\hadoop\sbin>hdfs dfs -put "C:\Users\PRANTO\Documents\Amazon Sale Report.csv" /project/  
  
C:\hadoop\sbin>hdfs dfs -ls /project  
Found 1 items  
-rw-r--r--  1 PRANTO supergroup  68923428 2026-09-03 23:37 /project/Amazon Sale Report.csv  
  
C:\hadoop\sbin>
```

4.2 Category-wise Total Sales

Calculates the total sales amount for each product category.

This analysis consists of:

1. **Driver Class:** Sets the configuration for Mapper and Reducer classes.
2. **Mapper Class:** Extracts the product category and sales amount from each record.
3. **Reducer Class:** Calculates the total sales amount for each category.

4.3 Driver Class

```
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class SalesDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {

            System.err.println(
                "Usage: SalesDriver <input.csv> <output>"
            );

            System.exit(2);
        }

        Configuration conf = new Configuration();

        Job job = Job.getInstance(
            conf,
            "Category Wise Total Sales"
        );

        job.setJarByClass(SalesDriver.class);

        job.setMapperClass(SalesMapper.class);
        job.setReducerClass(SalesReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(DoubleWritable.class);

        FileInputFormat.addInputPath(
            job,
            new Path(args[0])
        );

        FileOutputFormat.setOutputPath(
            job,
            new Path(args[1])
        );

        System.exit(
            job.waitForCompletion(true) ? 0 : 1
        );
    }
}
```

4.4 Mapper Class

```

import java.io.IOException;
import java.util.ArrayList;
import java.util.List;

import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class SalesMapper
    extends Mapper<LongWritable, Text, Text, DoubleWritable> {

    private int categoryIndex = -1;
    private int amountIndex = -1;

    private final Text outKey = new Text();
    private final DoubleWritable outValue = new DoubleWritable();

    private boolean headerProcessed = false;

    @Override
    public void map(LongWritable key,
                    Text value,
                    Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.trim().isEmpty()) {
            return;
        }

        // First line = CSV header
        if (!headerProcessed) {

            String[] header = parseCSV(line);

            for (int i = 0; i < header.length; i++) {

                String column = header[i]
                    .replace("\uFEFF", "")
                    .trim();

                if (column.equalsIgnoreCase("Category")) {
                    categoryIndex = i;
                }

                if (column.equalsIgnoreCase("Amount")) {
                    amountIndex = i;
                }
            }

            headerProcessed = true;
            return;
        }

        // Ignore if required columns are not found
        if (categoryIndex == -1 || amountIndex == -1) {
            return;
        }
    }

```

```

String[] data = parseCSV(line);

if (data.length <= categoryIndex ||
    data.length <= amountIndex) {
    return;
}

String category = data[categoryIndex].trim();
String amountText = data[amountIndex].trim();

if (category.isEmpty() || amountText.isEmpty()) {
    return;
}

try {
    double amount = Double.parseDouble(
        amountText.replace(",", ""));

    outKey.set(category);
    outValue.set(amount);

    context.write(outKey, outValue);
} catch (NumberFormatException e) {
    // Invalid amount is ignored
}
}

// Simple CSV parser supporting quoted commas
private String[] parseCSV(String line) {

    List<String> fields = new ArrayList<>();

    StringBuilder current = new StringBuilder();

    boolean inQuotes = false;

    for (int i = 0; i < line.length(); i++) {
        char c = line.charAt(i);

        if (c == '"') {
            if (inQuotes &&
                i + 1 < line.length() &&
                line.charAt(i + 1) == '"') {
                current.append('');
                i++;
            } else {
                inQuotes = !inQuotes;
            }
        } else if (c == ',' && !inQuotes) {

```



```

        fields.add(current.toString());
        current.setLength(0);

    } else {

        current.append(c);
    }
}

fields.add(current.toString());

return fields.toArray(new String[0]);
}
}

```

4.5 Reducer Class

```

import java.io.IOException;

import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class SalesReducer
    extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {

    private final DoubleWritable result = new DoubleWritable();

    @Override
    public void reduce(Text key,
                      Iterable<DoubleWritable> values,
                      Context context)
        throws IOException, InterruptedException {

        double sum = 0.0;

        for (DoubleWritable value : values) {
            sum += value.get();
        }

        result.set(sum);

        context.write(key, result);
    }
}

```

4.6 Execution

```
C:\Windows\System32>cd \
C:\>cd hadoop
C:\hadoop>cd sbin
C:\hadoop\sbin>start-dfs.cmd
C:\hadoop\sbin>start-yarn.cmd
starting yarn daemons

C:\hadoop\sbin>jps
16192 NodeManager
16272 DataNode
13544 NameNode
12700 Jps
15084 ResourceManager

C:\hadoop\sbin>cd C:\Users\PRANTO\Documents\Final Project
C:\Users\PRANTO\Documents\Final Project>mkdir category_sales
C:\Users\PRANTO\Documents\Final Project>cd category_sales
C:\Users\PRANTO\Documents\Final Project\category_sales>dir
Volume in drive C is OS
Volume Serial Number is DEF2-1384

Directory of C:\Users\PRANTO\Documents\Final Project\category_sales

04-Sep-26 12:07 PM <DIR>          .
04-Sep-26 12:07 PM <DIR>          ..
02-Sep-26 12:09 AM             1,302 SalesDriver.java
02-Sep-26 12:08 AM             3,248 SalesMapper.java
02-Sep-26 12:09 AM             709 SalesReducer.java
3 File(s)                    5,259 bytes
2 Dir(s)    225,256,652,000 bytes free

C:\Users\PRANTO\Documents\Final Project\category_sales>mkdir classes
C:\Users\PRANTO\Documents\Final Project\category_sales>for /f "delims=" %i in ('hadoop classpath') do set "HCP=%i"
C:\Users\PRANTO\Documents\Final Project\category_sales>set "HCP=C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\common;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\hdfs;C:\hadoop\share\hadoop\hdfs\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\yarn\lib\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\mapreduce\lib\*;C:\hadoop\share\hadoop\mapreduce\*"
C:\Users\PRANTO\Documents\Final Project\category_sales>javac -cp "%HCP%" -d classes *.java
C:\Users\PRANTO\Documents\Final Project\category_sales>jar -cvf CategorySales.jar -C classes .
added manifest
adding: SalesDriver.class(in = 1545) (out= 866)(deflated 43%)
adding: SalesMapper.class(in = 3288) (out= 1622)(deflated 50%)
adding: SalesReducer.class(in = 1713) (out= 713)(deflated 58%)

C:\Users\PRANTO\Documents\Final Project\category_sales>hadoop jar CategorySales.jar SalesDriver "/project/Amazon Sale Report.csv" /project/out_cs
2026-09-04 12:08:35,197 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 12:08:35,633 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-04 12:08:35,642 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/.staging/job_1788501357057_0002
2026-09-04 12:08:35,799 INFO input.FileInputFormat: Total input files to process : 1
2026-09-04 12:08:35,878 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-04 12:08:35,968 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788501357057_0002
2026-09-04 12:08:35,969 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-04 12:08:36,081 INFO conf.Configuration: resource-types.xml not found
2026-09-04 12:08:36,081 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-04 12:08:36,131 INFO impl.YarnClientImpl: Submitted application application_1788501357057_0002
2026-09-04 12:08:36,151 INFO mapreduce.Job: The url to track the job: http://DESKTOP-SMP:8088/proxy/application_1788501357057_0002/
2026-09-04 12:08:36,151 INFO mapreduce.Job: Running job: job_1788501357057_0002
2026-09-04 12:08:43,305 INFO mapreduce.Job: Job job_1788501357057_0002 running in uber mode : false
2026-09-04 12:08:43,311 INFO mapreduce.Job: map 0% reduce 0%
2026-09-04 12:08:48,368 INFO mapreduce.Job: map 100% reduce 0%
2026-09-04 12:08:54,438 INFO mapreduce.Job: map 100% reduce 100%
2026-09-04 12:08:54,450 INFO mapreduce.Job: Job job_1788501357057_0002 completed successfully
2026-09-04 12:08:54,518 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=1951062
    FILE: Number of bytes written=4379479
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=68923545
    HDFS: Number of bytes written=221
    HDFS: Number of read operations=8
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=1
    Launched reduce tasks=1
    Data-local map tasks=1
    Total time spent by all maps in occupied slots (ms)=2918
    Total time spent by all reduces in occupied slots (ms)=2469
    Total time spent by all map tasks (ms)=2918
    Total time spent by all reduce tasks (ms)=2469
    Total vcore-milliseonds taken by all map tasks=2918
    Total vcore-milliseonds taken by all reduce tasks=2469
    Total megabyte-milliseonds taken by all map tasks=2988032
    Total megabyte-milliseonds taken by all reduce tasks=2528256
```



```

Map-Reduce Framework
  Map input records=128976
  Map output records=121180
  Map output bytes=1708696
  Map output materialized bytes=1951062
  Input split bytes=117
  Combine input records=0
  Combine output records=0
  Reduce input groups=9
  Reduce shuffle bytes=1951062
  Reduce input records=121180
  Reduce output records=9
  Spilled Records=242360
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=81
  CPU time spent (ms)=2576
  Physical memory (bytes) snapshot=682975232
  Virtual memory (bytes) snapshot=984530944
  Total committed heap usage (bytes)=604504064
  Peak Map Physical memory (bytes)=459382784
  Peak Map Virtual memory (bytes)=614010880
  Peak Reduce Physical memory (bytes)=223592448
  Peak Reduce Virtual memory (bytes)=370520064
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=68923428
File Output Format Counters
  Bytes Written=221

```

4.7 Result

```

C:\Users\PRANTO\Documents\Final Project\category_sales>hdfs dfs -cat /project/out_cs/part-r-00000
Blouse 458408.18
Bottom 150667.97999999998
Dupatta 915.0
Ethnic Dress 791217.66000000001
Saree 123933.76000000001
Set 3.920412402999967E7
Top 5347792.300000001
Western Dress 1.1216072689999977E7
kurta 2.129954669999936E7

C:\Users\PRANTO\Documents\Final Project\category_sales>

```

4.8 Status-wise Order Count

Calculates the number of orders for each order status.

This analysis consists of:

1. **Driver Class:** Sets the configuration for Mapper and Reducer classes.
2. **Mapper Class:** Extracts the order status from each record and emits it with a count of 1.
3. **Reducer Class:** Calculates the total number of orders for each status.

4.9 Driver Class

```

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class StatusDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {

            System.err.println(
                "Usage: StatusDriver <input.csv> <output>"
            );

            System.exit(2);
        }

        Configuration conf = new Configuration();

        Job job = Job.getInstance(
            conf,
            "Status Wise Order Count"
        );

        job.setJarByClass(StatusDriver.class);

        job.setMapperClass(StatusMapper.class);
        job.setReducerClass(StatusReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(
            job,
            new Path(args[0])
        );

        FileOutputFormat.setOutputPath(
            job,
            new Path(args[1])
        );

        System.exit(
            job.waitForCompletion(true) ? 0 : 1
        );
    }
}

```

4.10 Mapper Class

```

import java.io.IOException;
import java.util.ArrayList;
import java.util.List;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class StatusMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {

    private int statusIndex = -1;
    private boolean headerProcessed = false;

    private final IntWritable one = new IntWritable(1);
    private final Text outKey = new Text();

    @Override
    public void map(LongWritable key,
                    Text value,
                    Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.trim().isEmpty()) {
            return;
        }

        // First line is the header
        if (!headerProcessed) {

            String[] header = parseCSV(line);

            for (int i = 0; i < header.length; i++) {

                String column = header[i]
                    .replace("\uFEFF", "")
                    .trim();

                if (column.equalsIgnoreCase("Status")) {
                    statusIndex = i;
                }
            }

            headerProcessed = true;
            return;
        }

        if (statusIndex == -1) {
            return;
        }

        String[] data = parseCSV(line);

        if (data.length <= statusIndex) {
            return;
        }
    }

```

```

    String status = data[statusIndex].trim();

    if (status.isEmpty()) {
        return;
    }

    outKey.set(status);

    context.write(outKey, one);
}

private String[] parseCSV(String line) {
    List<String> fields = new ArrayList<>();

    StringBuilder current = new StringBuilder();

    boolean inQuotes = false;

    for (int i = 0; i < line.length(); i++) {
        char c = line.charAt(i);

        if (c == '"') {
            if (inQuotes &&
                i + 1 < line.length() &&
                line.charAt(i + 1) == '"') {
                current.append('"');
                i++;
            } else {
                inQuotes = !inQuotes;
            }
        } else if (c == ',' && !inQuotes) {
            fields.add(current.toString().trim());
            current.setLength(0);
        } else {
            current.append(c);
        }
    }

    fields.add(current.toString().trim());

    return fields.toArray(new String[0]);
}

```

4.11 Reducer Class

```
import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class StatusReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    private final IntWritable result = new IntWritable();

    @Override
    public void reduce(Text key,
                      Iterable<IntWritable> values,
                      Context context)
        throws IOException, InterruptedException {

        int count = 0;

        for (IntWritable value : values) {
            count += value.get();
        }

        result.set(count);

        context.write(key, result);
    }
}
```

4.12 Execution

```
C:\Users\PRANTO\Documents\Final Project\category_sales>cd..
C:\Users\PRANTO\Documents\Final Project>mkdir status_count
C:\Users\PRANTO\Documents\Final Project>cd status_count
C:\Users\PRANTO\Documents\Final Project\status_count>dir
Volume in drive C is OS
Volume Serial Number is DEF2-1384

Directory of C:\Users\PRANTO\Documents\Final Project\status_count

04-Sep-26 12:45 PM <DIR>          .
04-Sep-26 12:45 PM <DIR>          ..
02-Sep-26 02:03 PM             1,299 StatusDriver.java
02-Sep-26 02:02 PM             2,569 StatusMapper.java
02-Sep-26 02:03 PM             698 StatusReducer.java
               3 File(s)         4,558 bytes
               2 Dir(s)        221,885,046,784 bytes free

C:\Users\PRANTO\Documents\Final Project\status_count>mkdir classes
C:\Users\PRANTO\Documents\Final Project\status_count>for /f "delims=" %i in ('hadoop classpath') do set "HCP=%i"

C:\Users\PRANTO\Documents\Final Project\status_count>set "HCP=C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\common\lib*;C:\hadoop\share\hadoop\common*;C:\hadoop\share\hadoop\hdfs;C:\hadoop\share\hadoop\hdfs\lib*;C:\hadoop\share\hadoop\hdfs*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\yarn\lib*;C:\hadoop\share\hadoop\yarn*;C:\hadoop\share\hadoop\mapreduce\lib*;C:\hadoop\share\hadoop\mapreduce**"

C:\Users\PRANTO\Documents\Final Project\status_count>javac -cp "%HCP%" -d classes *.java
C:\Users\PRANTO\Documents\Final Project\status_count>jar -cvf StatusCount.jar -C classes .
added manifest
adding: StatusDriver.class(in = 1545) (out= 862)(deflated 44%)
adding: StatusMapper.class(in = 2949) (out= 1422)(deflated 51%)
adding: StatusReducer.class(in = 1694) (out= 711)(deflated 58%)
```



```

C:\Users\PRANTO\Documents\Final Project\status_count>hadoop jar StatusCount.jar StatusDriver "/project/Amazon Sale Report.csv" /project/out_sc
2026-09-04 12:46:33,882 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 12:46:34,420 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-04 12:46:34,436 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/.staging/job_1788501357057_0003
2026-09-04 12:46:34,617 INFO input.FileInputFormat: Total input files to process : 1
2026-09-04 12:46:34,689 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-04 12:46:34,816 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788501357057_0003
2026-09-04 12:46:34,816 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-04 12:46:34,968 INFO conf.Configuration: resource-types.xml not found
2026-09-04 12:46:34,972 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-04 12:46:35,038 INFO impl.YarnClientImpl: Submitted application application_1788501357057_0003
2026-09-04 12:46:35,071 INFO mapreduce.Job: The url to track the job: http://DESKTOP-SMP:8088/proxy/application_1788501357057_0003/
2026-09-04 12:46:35,071 INFO mapreduce.Job: Running job: job_1788501357057_0003
2026-09-04 12:46:42,191 INFO mapreduce.Job: Job job_1788501357057_0003 running in uber mode : false
2026-09-04 12:46:42,191 INFO mapreduce.Job: map 0% reduce 0%
2026-09-04 12:46:48,284 INFO mapreduce.Job: map 100% reduce 0%
2026-09-04 12:46:53,349 INFO mapreduce.Job: map 100% reduce 100%
2026-09-04 12:46:53,364 INFO mapreduce.Job: Job job_1788501357057_0003 completed successfully
2026-09-04 12:46:53,430 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=2509523
  FILE: Number of bytes written=5496395
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=68923545
  HDFS: Number of bytes written=323
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=3232
  Total time spent by all reduces in occupied slots (ms)=2475
  Total time spent by all map tasks (ms)=3232
  Total time spent by all reduce tasks (ms)=2475
  Total vcore-milliseconds taken by all map tasks=3232
  Total vcore-milliseconds taken by all reduce tasks=2475
  Total megabyte-milliseconds taken by all map tasks=3309568
  Total megabyte-milliseconds taken by all reduce tasks=2534400

```

Map-Reduce Framework

```

Map input records=128976
Map output records=128975
Map output bytes=2251567
Map output materialized bytes=2509523
Input split bytes=117
Combine input records=0
Combine output records=0
Reduce input groups=13
Reduce shuffle bytes=2509523
Reduce input records=128975
Reduce output records=13
Spilled Records=257950
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=69
CPU time spent (ms)=2452
Physical memory (bytes) snapshot=707977216
Virtual memory (bytes) snapshot=974184448
Total committed heap usage (bytes)=585629696
Peak Map Physical memory (bytes)=491278336
Peak Map Virtual memory (bytes)=609714176
Peak Reduce Physical memory (bytes)=216698880
Peak Reduce Virtual memory (bytes)=364576768

```

Shuffle Errors

```

BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

```

File Input Format Counters

```
Bytes Read=68923428
```

File Output Format Counters

```
Bytes Written=323
```

4.13 Result

```
C:\Users\PRANTO\Documents\Final Project\status_count>hdfs dfs -cat /project/out_sc/part-r-00000
Cancelled          18332
Pending 658
Pending - Waiting for Pick Up    281
Shipped 77804
Shipped - Damaged          1
Shipped - Delivered to Buyer    28769
Shipped - Lost in Transit      5
Shipped - Out for Delivery     35
Shipped - Picked Up          973
Shipped - Rejected by Buyer     11
Shipped - Returned to Seller    1953
Shipped - Returning to Seller   145
Shipping            8

C:\Users\PRANTO\Documents\Final Project\status_count>
```

4.14 Fulfilment-wise Total Quantity

Calculates the total quantity of products for each fulfilment type.

This analysis consists of:

1. **Driver Class:** Sets the configuration for Mapper and Reducer classes.
2. **Mapper Class:** Extracts the fulfilment type and quantity from each record.
3. **Reducer Class:** Calculates the total quantity for each fulfilment type.

4.15 Driver Class

```

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class FulfilmentDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {

            System.err.println(
                "Usage: FulfilmentDriver <input.csv> <output>"
            );

            System.exit(2);
        }

        Configuration conf = new Configuration();

        Job job = Job.getInstance(
            conf,
            "Fulfilment Wise Total Quantity"
        );

        job.setJarByClass(FulfilmentDriver.class);

        job.setMapperClass(FulfilmentMapper.class);
        job.setReducerClass(FulfilmentReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(
            job,
            new Path(args[0])
        );

        FileOutputFormat.setOutputPath(
            job,
            new Path(args[1])
        );

        System.exit(
            job.waitForCompletion(true) ? 0 : 1
        );
    }
}

```

4.16 Mapper Class

```
import java.io.IOException;
import java.util.ArrayList;
import java.util.List;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class FulfilmentMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {

    private int fulfilmentIndex = -1;
    private int qtyIndex = -1;
    private boolean headerProcessed = false;

    private final Text outKey = new Text();
    private final IntWritable outValue = new IntWritable();

    @Override
    public void map(LongWritable key,
                   Text value,
                   Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.trim().isEmpty()) {
            return;
        }

        // Read header
        if (!headerProcessed) {

            String[] header = parseCSV(line);

            for (int i = 0; i < header.length; i++) {

                String column = header[i]
                    .replace("\uFEFF", "")
                    .trim();

                if (column.equalsIgnoreCase("Fulfilment")) {
                    fulfilmentIndex = i;
                }

                if (column.equalsIgnoreCase("Qty")) {
                    qtyIndex = i;
                }
            }
        }
    }
}
```

```

        headerProcessed = true;
        return;
    }

    if (fulfilmentIndex == -1 || qtyIndex == -1) {
        return;
    }

    String[] data = parseCSV(line);

    if (data.length <= Math.max(fulfilmentIndex, qtyIndex)) {
        return;
    }

    String fulfilment = data[fulfilmentIndex].trim();
    String qtyText = data[qtyIndex].trim();

    if (fulfilment.isEmpty() || qtyText.isEmpty()) {
        return;
    }

    try {
        int qty = Integer.parseInt(qtyText);

        outKey.set(fulfilment);
        outValue.set(qty);

        context.write(outKey, outValue);
    } catch (NumberFormatException e) {
        // Ignore invalid quantity
    }
}

private String[] parseCSV(String line) {
    List<String> fields = new ArrayList<>();
    StringBuilder current = new StringBuilder();

    boolean inQuotes = false;

    for (int i = 0; i < line.length(); i++) {
        char c = line.charAt(i);

        if (c == '"') {
            if (inQuotes &&
                i + 1 < line.length() &&
                line.charAt(i + 1) == '"') {
                current.append('');
                i++;
            } else {

```

```

        inQuotes = !inQuotes;
    }

    } else if (c == ',' && !inQuotes) {

        fields.add(current.toString().trim());
        current.setLength(0);

    } else {

        current.append(c);
    }
}

fields.add(current.toString().trim());

return fields.toArray(new String[0]);
}
}

```

4.17 Reducer Class

```

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class FulfilmentReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    private final IntWritable result = new IntWritable();

    @Override
    public void reduce(Text key,
        Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int total = 0;

        for (IntWritable value : values) {
            total += value.get();
        }

        result.set(total);

        context.write(key, result);
    }
}

```


4.18 Execution

```
C:\Users\PRANTO\Documents\Final Project\status_count>cd..

C:\Users\PRANTO\Documents\Final Project>mkdir fulfilment_quantity

C:\Users\PRANTO\Documents\Final Project>cd fulfilment_quantity

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>dir
Volume in drive C is OS
Volume Serial Number is DEF2-1384

Directory of C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity

04-Sep-26  01:25 PM    <DIR>          .
04-Sep-26  01:24 PM    <DIR>          ..
02-Sep-26  06:07 PM                1,326 FulfilmentDriver.java
02-Sep-26  06:07 PM                3,047 FulfilmentMapper.java
02-Sep-26  06:07 PM                694 FulfilmentReducer.java
               3 File(s)              5,067 bytes
               2 Dir(s)  221,738,532,864 bytes free

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>mkdir classes

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>for /f "delims=" %i in ('hadoop classpath') do set "HCP=%i"

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>set "HCP=C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\common;C:\hadoop\share\hadoop\common\lib*;C:\hadoop\share\hadoop\common*;C:\hadoop\share\hadoop\hdfs;C:\hadoop\share\hadoop\hdfs\lib*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\yarn\lib*;C:\hadoop\share\hadoop\yarn*;C:\hadoop\share\hadoop\mapreduce\lib*;C:\hadoop\share\hadoop\mapreduce*"

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>javac -cp "%HCP%" -d classes *.java

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>jar -cvf FulfilmentQuantity.jar -C classes .
added manifest
adding: FulfilmentDriver.class(in = 1572) (out= 871)(deflated 44%)
adding: FulfilmentMapper.class(in = 3299) (out= 1633)(deflated 50%)
adding: FulfilmentReducer.class(in = 1702) (out= 717)(deflated 57%)

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>hadoop jar FulfilmentQuantity.jar FulfilmentDriver "/project/Amazon Sale Report.csv" /project/out_fq
2026-09-04 13:26:46,469 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 13:26:46,898 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-04 13:26:46,912 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/.staging/job_1788501357057_0004
2026-09-04 13:26:47,093 INFO input.FileInputFormat: Total input files to process : 1
2026-09-04 13:26:47,150 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-04 13:26:47,251 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788501357057_0004
2026-09-04 13:26:47,255 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-04 13:26:47,356 INFO conf.Configuration: resource-types.xml not found
2026-09-04 13:26:47,356 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-04 13:26:47,431 INFO impl.YarnClientImpl: Submitted application application_1788501357057_0004
2026-09-04 13:26:47,451 INFO mapreduce.Job: The url to track the job: http://DESKTOP-SMP:8088/proxy/application_1788501357057_0004/
2026-09-04 13:26:47,451 INFO mapreduce.Job: Running job: job_1788501357057_0004
2026-09-04 13:26:54,593 INFO mapreduce.Job: Job job_1788501357057_0004 running in uber mode : false
2026-09-04 13:26:54,598 INFO mapreduce.Job:  map 0% reduce 0%
2026-09-04 13:26:59,668 INFO mapreduce.Job:  map 100% reduce 0%
2026-09-04 13:27:04,737 INFO mapreduce.Job:  map 100% reduce 100%
2026-09-04 13:27:05,758 INFO mapreduce.Job: Job job_1788501357057_0004 completed successfully
2026-09-04 13:27:05,829 INFO mapreduce.Job: Counters: 54
    File System Counters
        FILE: Number of bytes read=1755235
        FILE: Number of bytes written=3987849
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=68923545
        HDFS: Number of bytes written=28
        HDFS: Number of read operations=8
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=2
        HDFS: Number of bytes read erasure-coded=0
    Job Counters
        Launched map tasks=1
        Launched reduce tasks=1
        Data-local map tasks=1
        Total time spent by all maps in occupied slots (ms)=3123
        Total time spent by all reduces in occupied slots (ms)=2467
        Total time spent by all map tasks (ms)=3123
        Total time spent by all reduce tasks (ms)=2467
        Total vcore-milliseconds taken by all map tasks=3123
        Total vcore-milliseconds taken by all reduce tasks=2467
        Total megabyte-milliseconds taken by all map tasks=3197952
        Total megabyte-milliseconds taken by all reduce tasks=2526208
```

```

Map-Reduce Framework
  Map input records=128976
  Map output records=128975
  Map output bytes=1497279
  Map output materialized bytes=1755235
  Input split bytes=117
  Combine input records=0
  Combine output records=0
  Reduce input groups=2
  Reduce shuffle bytes=1755235
  Reduce input records=128975
  Reduce output records=2
  Spilled Records=257950
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=74
  CPU time spent (ms)=2513
  Physical memory (bytes) snapshot=692133888
  Virtual memory (bytes) snapshot=952705024
  Total committed heap usage (bytes)=536870912
  Peak Map Physical memory (bytes)=478662656
  Peak Map Virtual memory (bytes)=612007936
  Peak Reduce Physical memory (bytes)=213471232
  Peak Reduce Virtual memory (bytes)=340779008
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=68923428
File Output Format Counters
  Bytes Written=28

```

4.19 Result

```

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>hdfs dfs -cat /project/out_fq/part-r-00000
Amazon 84087
Merchant      32562

C:\Users\PRANTO\Documents\Final Project\fulfilment_quantity>

```


5. PIG ANALYSIS

Apache Pig is a high-level data flow platform used for processing and analyzing large datasets. It provides Pig Latin, a scripting language that simplifies the process of loading, transforming, grouping and analyzing data.

In this project, Apache Pig is used to analyze the Amazon Sale Report dataset. Five different analyses are performed:

✓ **Category-wise Total Quantity**

✓ **Category-wise Order Count**

✓ **Category-wise Average Amount**

✓ **Top 10 Categories**

✓ **Category-wise Maximum Amount**

The dataset is loaded and processed using Pig Latin scripts, and the analysis results are stored in HDFS for further viewing and evaluation.

5.1 Category-wise Total Quantity

Calculates the total quantity of products for each product category.

This analysis consists of:

1. **Pig Script:** Loads the Amazon Sale Report dataset, groups the records by product category and calculates the total quantity for each category.
2. **Execution:** Executes the Pig Latin script to process the dataset and generate the output.
3. **Result:** Displays the total quantity of products for each category.

5.2 Script

```

sales = LOAD '/project/Amazon Sale Report.csv'
USING PigStorage(',')
AS (
    index:int,
    order_id:chararray,
    date:chararray,
    status:chararray,
    fulfilment:chararray,
    sales_channel:chararray,
    ship_service_level:chararray,
    style:chararray,
    sku:chararray,
    category:chararray,
    size:chararray,
    asin:chararray,
    courier_status:chararray,
    qty:int,
    currency:chararray,
    amount:double,
    ship_city:chararray,
    ship_state:chararray,
    ship_postal_code:chararray,
    ship_country:chararray,
    promotion_ids:chararray,
    b2b:chararray,
    fulfilled_by:chararray,
    unnamed_22:chararray
);

grouped_data = GROUP sales BY category;

category_total_quantity = FOREACH grouped_data GENERATE
    group AS category,
    SUM(sales.qty) AS total_quantity;

STORE category_total_quantity
INTO '/CategoryWiseTotalQuantity.txt'
USING PigStorage(',');

```

```

2026-09-04 17:49:00,037 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at 0.0.0.0:8032
2026-09-04 17:49:00,039 [main] INFO org.apache.hadoop.mapred.client.ApplicationStateDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 17:49:00,058 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at 0.0.0.0:8032
2026-09-04 17:49:00,060 [main] INFO org.apache.hadoop.mapred.client.ApplicationStateDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 17:49:00,075 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at 0.0.0.0:8032
2026-09-04 17:49:00,086 [main] INFO org.apache.hadoop.mapred.client.ApplicationStateDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 17:49:00,100 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 3 time(s).
2026-09-04 17:49:00,106 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MapReduceLauncher - Success!
2026-09-04 17:49:00,127 [main] INFO org.apache.pig.Main - Pig script completed in 25 seconds and 158 milliseconds (25158 ms)

```

5.4 Result

```
C:\Users\PRANTO\Documents\Final Project\pig>hdfs dfs -cat /CategoryWiseTotalQuantity.txt/part-*
Set,45289
Top,9903
Saree,152
kurta,45045
Blouse,863
Bottom,398
Dupatta,3
Category,
Ethnic Dress,1053
Western Dress,13943

C:\Users\PRANTO\Documents\Final Project\pig>
```

5.5 Category-wise Order Count

Counts the number of orders for each product category.

This analysis consists of:

1. **Pig Script:** Loads the Amazon Sale Report dataset, groups the records by product category and counts the number of orders for each category.
2. **Execution:** Executes the Pig Latin script to process the dataset and generate the output.
3. **Result:** Displays the number of orders for each category.

5.6 Script

```

sales = LOAD '/project/Amazon Sale Report.csv'
USING PigStorage(',')
AS (
    index:int,
    order_id:chararray,
    date:chararray,
    status:chararray,
    fulfilment:chararray,
    sales_channel:chararray,
    ship_service_level:chararray,
    style:chararray,
    sku:chararray,
    category:chararray,
    size:chararray,
    asin:chararray,
    courier_status:chararray,
    qty:int,
    currency:chararray,
    amount:double,
    ship_city:chararray,
    ship_state:chararray,
    ship_postal_code:chararray,
    ship_country:chararray,
    promotion_ids:chararray,
    b2b:chararray,
    fulfilled_by:chararray,
    unnamed_22:chararray
);

grouped_data = GROUP sales BY category;

category_order_count = FOREACH grouped_data GENERATE
    group AS category,
    COUNT(sales) AS order_count;

STORE category_order_count
INTO '/CategoryWiseOrderCount.txt'
USING PigStorage(',');

```



```

Users\PRANTO\Documents\Final Project\pig> pig -x mapreduce -f categoryWiseOrderCount.pig
2026-09-04 15:11:26,557 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-04 15:11:26,557 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-04 15:11:26,557 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-04 15:11:26,715 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-04 15:11:26,715 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788513086708.log
2026-09-04 15:11:27,023 [main] INFO org.apache.pig.impl.util.Util$1 - Defining bootstrapper for class: C:\Users\PRANTO\AppData\Local\Temp\1788513086708\org.apache.pig.impl.util.Util$1.class
2026-09-04 15:11:27,068 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mpared_job_tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-04 15:11:27,068 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-04 15:11:27,482 [main] INFO org.apache.pig.bserver - Pig script ID for the session: PIG-categoryWiseOrderCount.pig-18276a2f-c818-40a6-a815-39035c4592d7
2026-09-04 15:11:27,482 [main] INFO org.apache.pig.bserver - ATIS is enabled since the yarn.timeline.service.enabled set to false
2026-09-04 15:11:27,482 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mpared_textoutputformat_separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-04 15:11:27,962 [main] INFO org.apache.pig.tools.pigstats.ScriptStats - Pig features used in the script: GROUP_BY
2026-09-04 15:11:27,989 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-04 15:11:28,008 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULE5_ENABLE] [addForEach, ColumnMajorKeyPrune, FileConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOpt
2026-09-04 15:11:28,008 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULE5_ENABLE] [addForEach, ColumnMajorKeyPrune, FileConstantCalculator, ForEachConstantCalculator, SplitFilter, StreamTypeCastInserter]]
2026-09-04 15:11:28,062 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-04 15:11:28,108 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCCompiler - File concatenation threshold: 100 optimistic: false
2026-09-04 15:11:28,119 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizer - Choosing to move algebraic forward to combiner
2026-09-04 15:11:28,134 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-04 15:11:28,134 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-04 15:11:28,190 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:11:28,357 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn_resource_manager_system_metrics_publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:11:28,357 [main] INFO org.apache.pig.tools.pigstats.mapreduce.HBScriptStats - Pig script settings are added to the job
2026-09-04 15:11:28,373 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mpared_job_reduce_markreset_buffer_percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-04 15:11:28,373 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mpared.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-04 15:11:28,373 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mpared_output_compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-04 15:11:28,373 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers
2026-09-04 15:11:28,373 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reduce estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-04 15:11:28,389 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=68923428
2026-09-04 15:11:28,389 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-04 15:11:28,389 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reducing tasks is not present. Instead, use mapreduce.job.reduce
2026-09-04 15:11:28,389 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-04 15:11:28,418 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mpared_submit_replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-04 15:11:28,597 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/0.18.0-core-h3.jar to DistributedCache through /tmp/temp-1629065196/tmp1974131344/pig-0.18.
2026-09-04 15:11:29,022 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automation-1.11-8.jar to DistributedCache through /tmp/temp-1629065196/tmp147315984/automat
2026-09-04 15:11:29,062 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp-1629065196/tmp-1518365977/antlr
2026-09-04 15:11:29,093 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp-1629065196/tmp-249194300/joda-ti
2026-09-04 15:11:29,125 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp-1629065196/tmp-969626
2026-09-04 15:11:29,125 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp-1629065196/tmp-969626
2026-09-04 15:11:29,157 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-04 15:11:29,162 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-04 15:11:29,164 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-04 15:11:29,164 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-04 15:11:29,216 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-04 15:11:29,220 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:11:29,233 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mpared_task_id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-04 15:11:29,504 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/staging/job_1788511623088_0002
2026-09-04 15:11:29,519 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobSetJar(String).
2026-09-04 15:11:29,553 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigStorageFormat
2026-09-04 15:11:29,553 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapReduceUtil - Total input files to process : 1
2026-09-04 15:11:29,610 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapReduceUtil - Total input paths to process : 1
2026-09-04 15:11:29,610 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapReduceUtil - Total input paths (combined) to process : 1
2026-09-04 15:11:29,627 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobSubmitter - number of splits: 1
2026-09-04 15:11:29,709 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn_resource_manager_system_metrics_publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:11:29,803 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1788511623088_0002
2026-09-04 15:11:29,884 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-04 15:11:29,884 [JobControl] INFO org.apache.hadoop.mapreduce.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-04 15:11:29,935 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-04 15:11:29,935 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-04 15:11:29,978 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1788511623088_0002
2026-09-04 15:11:30,017 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - http://DESKTOP-SMP-8888/proxy/application_1788511623088_0002/
2026-09-04 15:11:30,017 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1788511623088_0002
2026-09-04 15:11:30,017 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases category_order_count,grouped_data,sales
2026-09-04 15:11:30,017 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: sales[1,8],sales[1,-1],category_order_count[32,23],grouped_data[30,15] C: category_order_count[
2026-09-04 15:11:30,026 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-04 15:11:30,026 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1788511623088_0002]
2026-09-04 15:11:42,281 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-09-04 15:11:42,281 [main] INFO org.apache.pig.backend.hadoop
```

5.8 Result

```
C:\Users\PRANTO\Documents\Final Project\pig>hdfs dfs -cat /CategoryWiseOrderCount.txt/part-*
Set,50284
Top,10622
Saree,164
kurta,49877
Blouse,926
Bottom,440
Dupatta,3
Category,0
Ethnic Dress,1159
Western Dress,15500

C:\Users\PRANTO\Documents\Final Project\pig>
```

5.9 Category-wise Average Amount

Calculates the average sales amount for each product category.

This analysis consists of:

1. **Pig Script:** Loads the Amazon Sale Report dataset, groups the records by product category and calculates the average sales amount for each category.
2. **Execution:** Executes the Pig Latin script to process the dataset and generate the output.
3. **Result:** Displays the average sales amount for each category.

5.10 Script

```

sales = LOAD '/project/Amazon Sale Report.csv'
USING PigStorage(',')
AS (
    index:int,
    order_id:chararray,
    date:chararray,
    status:chararray,
    fulfilment:chararray,
    sales_channel:chararray,
    ship_service_level:chararray,
    style:chararray,
    sku:chararray,
    category:chararray,
    size:chararray,
    asin:chararray,
    courier_status:chararray,
    qty:int,
    currency:chararray,
    amount:double,
    ship_city:chararray,
    ship_state:chararray,
    ship_postal_code:chararray,
    ship_country:chararray,
    promotion_ids:chararray,
    b2b:chararray,
    fulfilled_by:chararray,
    unnamed_22:chararray
);

grouped_data = GROUP sales BY category;

category_average_amount = FOREACH grouped_data GENERATE
    group AS category,
    AVG(sales.amount) AS average_amount;

STORE category_average_amount
INTO '/CategoryWiseAverageAmount.txt'
USING PigStorage(',');

```



```
Users\PRANTO/Documents/Final Project/pigpigp -x mapreduce -f categoryWiseAverageAmount.pig
2026-09-04 15:36:34,895 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-04 15:36:34,895 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-04 15:36:34,895 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExeType
2026-09-04 15:36:34,230 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
org.apache.pig.Main - Logging error messages to C:\hadooplogs\pig_1788514594230.log
2026-09-04 15:36:34,261 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\PRANTO\AppData\Local\Packages\Microsoft.Windows.Common-Infrastructure_9595b641-fa67-4887-a631-e6d40acbbcf7_xwwzrhqshwcpn!bootstrap not found
2026-09-04 15:36:34,261 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mappend.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-04 15:36:34,544 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-04 15:36:34,884 [main] INFO org.apache.pig.BFServeer - Pig Script ID for the session: PIG-categoryWiseAverageAmount.pig-f2fcab5a-d11e-47da-9886-0bf5deaf7fcc
2026-09-04 15:36:34,884 [WARN] org.apache.pig.toolservicer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-04 15:36:35,313 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapped.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-04 15:36:35,327 [main] INFO org.apache.pig.tools.pigstats.ScriptStats - Pig features used in the script: GROUP_BY
2026-09-04 15:36:35,342 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schemautils] was not set...will not generate code.
2026-09-04 15:36:35,364 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULES.ENLABLED=AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizeFilter, MergeJoinPushdown, NestedInjoinOptimizer, PartitionFilterOptimizer, PredicatePushDownOptimizer, PushDownOnEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamYieldInsertion]
2026-09-04 15:36:35,488 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-04 15:36:35,442 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MRCompiler - File concatenation thresholds: 100 optimistic? false
2026-09-04 15:36:35,451 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic forward to combiner
2026-09-04 15:36:35,462 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-04 15:36:35,462 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-04 15:36:35,512 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:36:35,691 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:36:35,691 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - MapReduce task settings are added to the job
2026-09-04 15:36:35,710 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mappend.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-04 15:36:35,710 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - mappend.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-04 15:36:35,712 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapped.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-04 15:36:35,712 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.InputSizeReducerEstimator
2026-09-04 15:36:35,732 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.InputSizeReducerEstimator - BytesPerReducer=100000000 maxReducers=999 totalInputFileSize=68923428
2026-09-04 15:36:35,732 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer - Setting Parallelism to 1
2026-09-04 15:36:35,732 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapped.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-04 15:36:35,732 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - This Job cannot be converted run in-process
2026-09-04 15:36:35,750 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapped.submit.replication is deprecated. Instead, use mapreduce.cluster.submit.file.replication
2026-09-04 15:36:35,985 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - Added jar file:C:/pig/lib/hado-core-h3.jar to DistributedCache through /tmp/tmp1425990574/tmp5E8031974/pig-0.18.0-rc1-jar
2026-09-04 15:36:35,937 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - Added jar file:C:/pig/lib/automation-1.11.8.jar to DistributedCache through /tmp/tmp1425990574/tmp-712639736/automato
n-1.11.8.jar
2026-09-04 15:36:35,968 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - Added jar file:C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/tmp1425990574/tmp327385634/antlr-run
time-3.4.jar
2026-09-04 15:36:36,000 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - Added jar file:C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/tmp1425990574/tmp-214714677/joda-tim
e-2.14.0.jar
2026-09-04 15:36:36,032 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.JobControlCompiler - Added jar file:C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/tmp1425990574/tmp3427748
RoaringBitmap-shaded-0.7.14.jar
2026-09-04 15:36:36,074 [main] INFO org.apache.pig.data.SchemaTuplesFrontend - key [mapreduce.task.attemptid] is missing from schema. Will not generate code.
2026-09-04 15:36:36,081 [main] INFO org.apache.pig.data.SchemaTuplesFrontend - Starting process to move generated code to distributed cache
2026-09-04 15:36:36,081 [main] INFO org.apache.pig.data.SchemaTuplesFrontend - Setting key [mapreduce.task.attemptid] with classes to deserialize []
2026-09-04 15:36:36,137 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MapReduceLauncher - i map-reduce job(s) waiting for submission.
2026-09-04 15:36:36,145 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:36:36,165 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mappend.task.id is deprecated. Instead, use mapreduce.task.attemptid
2026-09-04 15:36:36,409 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/.staging/job_1788511623088_0003
2026-09-04 15:36:36,427 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2026-09-04 15:36:36,464 [JobControl] INFO org.apache.pig.builtins.PigStorage - Using PigTextOutputFormat
2026-09-04 15:36:36,464 [JobControl] INFO org.apache.pig.backend.mapreduce.lib.input.FileInputSplitMapper - Total input files to process : 1
2026-09-04 15:36:36,464 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-04 15:36:36,541 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-04 15:36:36,583 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.Splitter - Number of splits: 1
2026-09-04 15:36:36,639 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:36:36,716 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1788511623088_0003
2026-09-04 15:36:36,718 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-04 15:36:36,808 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer - The url to track the job: http://DESKTOP-SWP-008B/proxy/application_1788511623088_0003/
2026-09-04 15:36:36,855 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-04 15:36:36,855 [JobControl] INFO org.apache.hadoop.yarn.api.resource.ResourceUtils - Unable to find "resource-types.xml".
2026-09-04 15:36:36,919 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1788511623088_0003
2026-09-04 15:36:36,962 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 3 time(s).
2026-09-04 15:36:36,962 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer - Detailed locations: M: sales[1,0],sales[-1,-1],category_average_amount[grouped_data,sales] C: category_average_a
mount[grouped_data[sales[1,0]],grouped_data[sales[-1,-1]]]
2026-09-04 15:36:36,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MapReduceLauncher - % complete
2026-09-04 15:36:36,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MapReduceLauncher - Running jobs are [job_1788511623088_0003]
2026-09-04 15:36:36,971 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MapReduceLauncher - SOK complete
2026-09-04 15:36:36,971 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MapReduceLauncher - Running jobs are [job_1788511623088_0003]
2026-09-04 15:36:36,971 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReducerLayer.MapReduceLauncher - Running jobs are [job_1788511623088_0003]
2026-09-04 15:36:37,384 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:36:37,392 [main] INFO org.apache.hadoop.mapped.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:36:37,392 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:3
```

5.12 Result

```
C:\Users\PRANTO\Documents\Final Project\pig>hdfs dfs -cat /CategoryWiseAverageAmount.txt/part-*
Set,833.3855709791216
Top,526.0986030496805
Saree,799.5726451612903
kurta,455.92710790504407
Blouse,520.3271055618615
Bottom,358.73328571428567
Dupatta,305.0
Category,
Ethnic Dress,723.8953888380605
Western Dress,762.7905801142524
```

5.13 Top 10 Categories

Identifies the top 10 product categories based on total quantity.

This analysis consists of:

1. **Pig Script:** Loads the Amazon Sale Report dataset, groups the records by product category, calculates the total quantity for each category and identifies the top 10 categories based on total quantity.
2. **Execution:** Executes the Pig Latin script to process the dataset and generate the output.
3. **Result:** Displays the top 10 product categories based on total quantity.

5.14 Script

```

sales = LOAD '/project/Amazon Sale Report.csv'
USING PigStorage(',')
AS (
    index:int,
    order_id:chararray,
    date:chararray,
    status:chararray,
    fulfilment:chararray,
    sales_channel:chararray,
    ship_service_level:chararray,
    style:chararray,
    sku:chararray,
    category:chararray,
    size:chararray,
    asin:chararray,
    courier_status:chararray,
    qty:int,
    currency:chararray,
    amount:double,
    ship_city:chararray,
    ship_state:chararray,
    ship_postal_code:chararray,
    ship_country:chararray,
    promotion_ids:chararray,
    b2b:chararray,
    fulfilled_by:chararray,
    unnamed_22:chararray
);

grouped_data = GROUP sales BY category;

category_quantity = FOREACH grouped_data GENERATE
    group AS category,
    SUM(sales.qty) AS total_quantity;

ordered_data = ORDER category_quantity BY total_quantity DESC;

top10_categories = LIMIT ordered_data 10;

STORE top10_categories
INTO '/Top10Categories.txt'
USING PigStorage(',');

```

5.15 Execution

```
C:\Users\PRANTO\Documents\Final Project>pig>pig -x mapreduce -f top10Categories.pig
2026-09-04 15:55:30,190 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-04 15:55:30,191 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-04 15:55:30,191 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-04 15:55:30,337 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-04 15:55:30,337 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788515730333.log
2026-09-04 15:55:30,695 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\PRANTO\.pigbootstrap not found
2026-09-04 15:55:30,645 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-04 15:55:30,645 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-04 15:55:30,959 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10Categories.pig-ebbfdda-eab2-4b74-9866-5cd88742501
2026-09-04 15:55:30,959 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-04 15:55:31,388 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-04 15:55:31,481 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,LIMIT
2026-09-04 15:55:31,416 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-04 15:55:31,429 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOpt
mizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]]
2026-09-04 15:55:31,471 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 690400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-04 15:55:31,593 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-04 15:55:31,520 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-04 15:55:31,551 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-91
2026-09-04 15:55:31,594 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-04 15:55:31,561 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-04 15:55:31,614 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:55:31,749 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:55:31,774 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-04 15:55:31,782 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-04 15:55:31,785 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-04 15:55:31,790 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutformat.compress
2026-09-04 15:55:31,794 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Starting process to move generated code to distributed cache
2026-09-04 15:55:31,799 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-04 15:55:31,805 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=68923428
2026-09-04 15:55:31,885 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-04 15:55:31,895 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Instead of reduce tasks is deprecated. Instead, use mapreduce.job.reducers
2026-09-04 15:55:31,805 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-04 15:55:31,835 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-04 15:55:31,895 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp1312045351/tmp1896706082/pig-0.18.0-core-h3.jar
2026-09-04 15:55:32,032 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp1312045351/tmp654765537/automaton-1.11-8.jar
2026-09-04 15:55:32,071 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp1312045351/tmp539909413/antlr-runtime-3.4.jar
2026-09-04 15:55:32,113 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp1312045351/tmp1436715240/joda-time-2.14.0.jar
2026-09-04 15:55:32,155 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp1312045351/tmp1690543653/RoaringBitmap-shaded-0.7.14.jar

2026-09-04 15:55:32,207 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-04 15:55:32,218 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] is false, will not generate code.
2026-09-04 15:55:32,218 [main] INFO org.apache.pig.data.SchemaTupleBackend - Starting process to move generated code to distributed cache
2026-09-04 15:55:32,218 [main] INFO org.apache.pig.data.SchemaTupleBackend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-04 15:55:32,326 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-04 15:55:32,334 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:55:32,356 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-04 15:55:32,600 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/staging/job_1788511623088_0008
2026-09-04 15:55:32,614 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job file set. User classes may not be found. See Job or JobsetJar(String).
2026-09-04 15:55:32,647 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-04 15:55:32,704 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.FileInputFormat - Total input files to process : 1
2026-09-04 15:55:32,704 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-04 15:55:32,722 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-04 15:55:32,784 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-04 15:55:32,823 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:55:32,898 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1788511623088_0008
2026-09-04 15:55:32,901 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-04 15:55:32,990 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-04 15:55:33,032 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-04 15:55:33,032 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-04 15:55:33,291 [JobControl] INFO org.apache.hadoop.yarn.client.impl.YarnClientImpl - Submitted application application_1788511623088_0008
2026-09-04 15:55:33,339 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - The url to track the job: http://DESKTOP-SMP:8088/proxy/application_1788511623088_0008/
2026-09-04 15:55:33,339 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - HadoopJobId: job_1788511623088_0008
2026-09-04 15:55:33,341 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases category,quantity,grouped_data,sales
2026-09-04 15:55:33,341 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: sales[1,8],sales[-1,-1],category,quantity[32,20],grouped_data[30,15] C: category,quantity[32,20]
2026-09-04 15:55:33,364 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-04 15:55:33,364 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1788511623088_0008]
2026-09-04 15:55:45,596 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 12% complete
2026-09-04 15:55:45,602 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1788511623088_0008]
2026-09-04 15:55:50,672 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 25% complete
2026-09-04 15:55:50,676 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1788511623088_0008]
2026-09-04 15:55:53,759 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:55:53,776 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:55:53,898 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:55:53,908 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:55:53,938 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:55:53,940 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:55:53,983 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-04 15:55:53,983 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-04 15:55:53,985 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-04 15:55:53,985 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-04 15:55:54,002 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=149
2026-09-04 15:55:54,002 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-04 15:55:54,002 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-04 15:55:54,041 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp1312045351/tmp635891879/pig-0.18.0-core-h3.jar
2026-09-04 15:55:54,072 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp1312045351/tmp-1562168748/automaton-1.11-8.jar
2026-09-04 15:55:54,104 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp1312045351/tmp541220863/antlr-runtime-3.4.jar
2026-09-04 15:55:54,135 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp1312045351/tmp-1211784547/joda-time-2.14.0.jar
2026-09-04 15:55:54,167 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp1312045351/tmp-264900420/RoaringBitmap-shaded-0.7.14.jar
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2026-09-04 15:55:54,177 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-04 15:55:54,177 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-04 15:55:54,177 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-04 15:55:54,177 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-04 15:55:54,198 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-04 15:55:54,198 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:55:54,283 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:55:54,214 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/job_178851623088_0009
2026-09-04 15:55:54,214 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobsetJar(String).
2026-09-04 15:55:54,246 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-04 15:55:54,246 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-04 15:55:54,246 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-04 15:55:54,294 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-04 15:55:54,341 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_178851623088_0009
2026-09-04 15:55:54,341 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-04 15:55:54,349 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-04 15:55:54,582 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_178851623088_0009
2026-09-04 15:55:54,597 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://DESKTOP-SMP:8088/proxy/application_178851623088_0009/
2026-09-04 15:55:54,710 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_178851623088_0009
2026-09-04 15:55:54,710 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases ordered_data
2026-09-04 15:55:54,710 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: ordered_data[36,15] C: R:
2026-09-04 15:56:07,099 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 37% complete
2026-09-04 15:56:07,105 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_178851623088_0009]
2026-09-04 15:56:13,716 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 64% complete
2026-09-04 15:56:13,724 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_178851623088_0009]
2026-09-04 15:56:14,754 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:14,764 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:56:14,764 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:14,857 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:56:14,872 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:14,882 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:56:14,900 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-04 15:56:14,900 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - mapped.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-04 15:56:14,900 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-04 15:56:14,900 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-04 15:56:14,900 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-04 15:56:14,951 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/0.18.0-core-h3.jar to DistributedCache through /tmp/temp1312045351/tmp-118003593/pig-0.18.0-core-h3.jar
2026-09-04 15:56:14,983 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp1312045351/tmp-1915165292/automato
n-1.11-8.jar
2026-09-04 15:56:14,999 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp1312045351/tmp107240008/antlr-ru
ntime-3.4.jar
2026-09-04 15:56:15,038 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp1312045351/tmp520218169/joda-time
-2.14.0.jar
2026-09-04 15:56:15,062 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp1312045351/tmp-1180976
256/RoaringBitmap-shaded-0.7.14.jar

2026-09-04 15:56:15,074 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-04 15:56:15,074 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-04 15:56:15,074 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-04 15:56:15,074 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-04 15:56:15,100 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-04 15:56:15,113 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:15,120 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:56:15,135 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/staging/job_178851623088_0010
2026-09-04 15:56:15,135 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobsetJar(String).
2026-09-04 15:56:15,162 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-04 15:56:15,162 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-04 15:56:15,164 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-04 15:56:15,204 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-04 15:56:15,242 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_178851623088_0010
2026-09-04 15:56:15,242 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-04 15:56:15,251 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-04 15:56:15,283 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_178851623088_0010
2026-09-04 15:56:15,283 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - The url to track the job: http://DESKTOP-SMP:8088/proxy/application_178851623088_0010/
2026-09-04 15:56:15,622 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_178851623088_0010
2026-09-04 15:56:15,630 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases ordered_data
2026-09-04 15:56:15,630 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: ordered_data[36,15] C: R:
2026-09-04 15:56:29,871 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 62% complete
2026-09-04 15:56:29,871 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_178851623088_0010]
2026-09-04 15:56:35,456 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 75% complete
2026-09-04 15:56:35,456 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_178851623088_0010]
2026-09-04 15:56:40,526 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:40,532 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:56:40,609 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:40,609 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:56:40,643 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:40,664 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-04 15:56:40,664 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapped.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-04 15:56:40,673 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-04 15:56:40,673 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-04 15:56:40,673 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-04 15:56:40,728 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/0.18.0-core-h3.jar to DistributedCache through /tmp/temp1312045351/tmp852842502/pig-0.18.0-
core-h3.jar
2026-09-04 15:56:40,752 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp1312045351/tmp1726452439/automato
n-1.11-8.jar
2026-09-04 15:56:40,768 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp1312045351/tmp583701782/antlr-ru
ntime-3.4.jar
2026-09-04 15:56:40,800 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp1312045351/tmp-1237353744/joda-ti
me-2.14.0.jar
2026-09-04 15:56:40,831 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp1312045351/tmp-5367201
89/RoaringBitmap-shaded-0.7.14.jar

2026-09-04 15:56:40,844 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-04 15:56:40,844 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-04 15:56:40,844 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-04 15:56:40,844 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-04 15:56:40,847 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-04 15:56:40,847 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:56:40,863 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-04 15:56:40,877 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/staging/job_178851623088_0011
2026-09-04 15:56:40,881 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobsetJar(String).
2026-09-04 15:56:40,897 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-04 15:56:40,897 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-04 15:56:40,897 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-04 15:56:40,942 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-04 15:56:40,985 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_178851623088_0011
2026-09-04 15:56:40,985 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-04 15:56:40,990 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-04 15:56:41,249 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_178851623088_0011
2026-09-04 15:56:41,254 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://DESKTOP-SMP:8088/proxy/application_178851623088_0011/
2026-09-04 15:56:41,356 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_178851623088_0011
2026-09-04 15:56:41,356 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases ordered_data
2026-09-04 15:56:41,356 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: ordered_data[36,15] C: R:
2026-09-04 15:56:55,618 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 87% complete
2026-09-04 15:56:55,618 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_178851623088_0011]
2026-09-04 15:57:01,700 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_178851623088_0011]
2026-09-04 15:57:06,771 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:06,797 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:06,892 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:06,892 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:06,921 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:06,924 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:06,945 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-04 15:57:06,969 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

```



```
HadoopVersion  PigVersion  UserId StartedAt FinishedAt Features
3.2.4 0.18.0 PRANTO 2026-09-04 15:55:31 2026-09-04 15:57:06 GROUP_BY,ORDER_BY,LIMIT
```

Success!

Job Stats (time in seconds):

JobId	Maps	Reduces	MaxMapTime	MinMapTime	AvgMapTime	MedianMapTime	MaxReduceTime	MinReduceTime	AvgReduceTime	MedianReductime	Alias	Feature Outputs
job_1788511623088_0008	1	1	3	3	3	2	2	2	2	2		category_quantity,grouped_data,sales
job_1788511623088_0009	1	1	2	2	2	2	2	2	2	2		ordered_data
job_1788511623088_0010	1	1	2	2	2	3	3	3	3	3		ordered_data
job_1788511623088_0011	1	1	3	3	3	3	3	3	3	3		ordered_data

Input(s):
Successfully read 128976 records (68923807 bytes) from: "/project/Amazon Sale Report.csv"

Output(s):
Successfully stored 10 records (121 bytes) in: "/Top10Categories.txt"

Counters:
Total records written : 10
Total bytes written : 121
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1788511623088_0008 --> job_1788511623088_0009,
job_1788511623088_0009 --> job_1788511623088_0010,
job_1788511623088_0010 --> job_1788511623088_0011,
job_1788511623088_0011

```
2026-09-04 15:57:06,973 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:06,987 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,016 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,019 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,052 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,054 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,089 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,095 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,111 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,111 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,139 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,146 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,178 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,188 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,221 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,231 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,266 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,274 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,316 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,335 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,373 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,385 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,419 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-04 15:57:07,425 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-04 15:57:07,449 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 3 time(s).
2026-09-04 15:57:07,449 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-04 15:57:07,490 [main] INFO org.apache.pig.Main - Pig script completed in 1 minute, 37 seconds and 328 milliseconds (97328 ms)
```

5.16 Result

```
C:\Users\PRANTO\Documents\Final Project\pig>hdfs dfs -cat /Top10Categories.txt/part-*
Set,45289
kurta,45045
Western Dress,13943
Top,9903
Ethnic Dress,1053
Blouse,863
Bottom,398
Saree,152
Dupatta,3
Category,
```

```
C:\Users\PRANTO\Documents\Final Project\pig>
```


5.17 Category-wise Maximum Amount

Finds the maximum sales amount for each product category.

This analysis consists of:

1. **Pig Script:** Loads the Amazon Sale Report dataset, groups the records by product category and finds the maximum sales amount for each category.
2. **Execution:** Executes the Pig Latin script to process the dataset and generate the output.
3. **Result:** Displays the maximum sales amount for each category.

5.18 Script

```

sales = LOAD '/project/Amazon Sale Report.csv'
USING PigStorage(',')
AS (
    index:int,
    order_id:chararray,
    date:chararray,
    status:chararray,
    fulfilment:chararray,
    sales_channel:chararray,
    ship_service_level:chararray,
    style:chararray,
    sku:chararray,
    category:chararray,
    size:chararray,
    asin:chararray,
    courier_status:chararray,
    qty:int,
    currency:chararray,
    amount:double,
    ship_city:chararray,
    ship_state:chararray,
    ship_postal_code:chararray,
    ship_country:chararray,
    promotion_ids:chararray,
    b2b:chararray,
    fulfilled_by:chararray,
    unnamed_22:chararray
);

grouped_data = GROUP sales BY category;

category_maximum_amount = FOREACH grouped_data GENERATE
    group AS category,
    MAX(sales.amount) AS max_amount;

STORE category_maximum_amount
INTO '/CategoryWiseMaximumAmount.txt'
USING PigStorage(',');

```

5.19 Execution

```
C:\Users\PRANTO\Documents\Final Project\pig>pig -x mapreduce -f categoryWiseMaximumAmount.pig
2026-09-05 19:01:12,233 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-05 19:01:12,233 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-05 19:01:12,373 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-05 19:01:12,373 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788613272363.log
2026-09-05 19:01:12,631 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\PRANTO\pigbootstrap not found
2026-09-05 19:01:12,668 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-05 19:01:12,668 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-05 19:01:12,992 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-categoryWiseMaximumAmount.pig-900003b3-9d40-4e0c-b4fb-2d58f9b5a643
2026-09-05 19:01:13,027 [main] WARN org.apache.pig.PigServer - AT5 is disabled since yarn.timeline-service.enabled set to false
2026-09-05 19:01:13,455 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-05 19:01:13,463 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2026-09-05 19:01:13,485 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-05 19:01:13,511 [main] INFO org.apache.pig.neplan.logical.optimizer.LogicalPlanOptimizer - {RULE5_ENABLED=[AddSortEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOpt
mizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedInitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-05 19:01:13,548 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-05 19:01:13,587 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-05 19:01:13,603 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-05 19:01:13,627 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-05 19:01:13,627 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-05 19:01:13,681 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:13,842 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-05 19:01:13,858 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-05 19:01:13,862 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-05 19:01:13,864 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-05 19:01:13,864 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-05 19:01:13,868 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-05 19:01:13,868 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-05 19:01:13,880 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=68923428
2026-09-05 19:01:13,880 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-05 19:01:13,880 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapreduce.task.io.sort.mb is deprecated. Instead, use mapreduce.job.reduces
2026-09-05 19:01:13,880 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-05 19:01:13,896 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-05 19:01:14,591 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp-835274160/tmp-1778313711/pig-0.18.0-core-h3.jar
2026-09-05 19:01:14,623 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp-835274160/tmp1246655577/automato
n-1.11-8.jar
2026-09-05 19:01:14,668 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp-835274160/tmp-1815474814/antlr-
runtime-3.4.jar
2026-09-05 19:01:14,707 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp-835274160/tmp395230453/joda-time
-2.14.0.jar
2026-09-05 19:01:14,740 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp-835274160/tmp-1061597
130/RoaringBitmap-shaded-0.7.14.jar

2026-09-05 19:01:14,768 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-05 19:01:14,774 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-05 19:01:14,774 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-05 19:01:14,774 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-05 19:01:14,837 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-05 19:01:14,845 [JobControl] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:14,866 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-05 19:01:15,101 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/PRANTO/job_1788613192920_0001
2026-09-05 19:01:15,115 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobResJar(String).
2026-09-05 19:01:15,148 [JobControl] INFO org.apache.pig.builtins.PigStorage - Using PigTextInputFormat
2026-09-05 19:01:15,152 [JobControl] INFO org.apache.pig.backend.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-05 19:01:15,152 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-05 19:01:15,152 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-05 19:01:15,291 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-05 19:01:15,324 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-05 19:01:15,374 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1788613192920_0001
2026-09-05 19:01:15,435 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-05 19:01:15,525 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-05 19:01:15,574 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-05 19:01:15,574 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-05 19:01:15,990 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1788613192920_0001
2026-09-05 19:01:16,029 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://DESKTOP-SMP:8088/proxy/application_1788613192920_0001/
2026-09-05 19:01:16,029 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1788613192920_0001
2026-09-05 19:01:16,029 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases category_maximum_amount,grouped_data,sales
2026-09-05 19:01:16,029 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: sales[1,8],sales[-1,-1],category_maximum_amount[32,26],grouped_data[30,15] C: category_maxim_m
ount[32,26],grouped_data[30,15] R: category_maximum_amount[32,26]
2026-09-05 19:01:16,044 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-05 19:01:16,044 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1788613192920_0001]
2026-09-05 19:01:31,371 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-09-05 19:01:31,371 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1788613192920_0001]
2026-09-05 19:01:31,514 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1788613192920_0001]
2026-09-05 19:01:41,590 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:41,599 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-05 19:01:42,500 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:42,516 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-05 19:01:42,549 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:42,553 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-05 19:01:42,647 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-05 19:01:42,657 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion PigVersion UserId StartedAt FinishedAt Features
3.2.4 0.18.0 PRANTO 2026-09-05 19:01:13 2026-09-05 19:01:42 GROUP_BY

Success!

Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReduceTime Alias Feature Outputs
job_1788613192920_0001 1 1 4 4 4 3 3 3 3 category_maximum_amount,grouped_data,sales GROUP_BY,COMBINER /CategoryWiseMaximumAmount.txt,

Input(s):
Successfully read 128976 records (68923807 bytes) from: "/project/Amazon Sale Report.csv"

Output(s):
Successfully stored 10 records (143 bytes) in: "/CategoryWiseMaximumAmount.txt"

Counters:
Total records written : 10
Total bytes written : 143
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1788613192920_0001

2026-09-05 19:01:42,663 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:42,673 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-05 19:01:42,706 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:42,712 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-05 19:01:42,738 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-05 19:01:42,742 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-05 19:01:42,766 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 3 time(s).
2026-09-05 19:01:42,768 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-05 19:01:42,785 [main] INFO org.apache.pig.Main - Pig script completed in 30 seconds and 602 milliseconds (30602 ms)
```

5.20 Result

```
C:\Users\PRANTO\Documents\Final Project\pig>hdfs dfs -cat /CategoryWiseMaximumAmount.txt/part-*  
Set,5584.0  
Top,1797.0  
Saree,2058.0  
kurta,2796.0  
Blouse,1266.66  
Bottom,1028.58  
Dupatta,305.0  
Category,  
Ethnic Dress,1449.0  
Western Dress,2860.0  
  
C:\Users\PRANTO\Documents\Final Project\pig>
```

6. PROJECT REPOSITORY

<https://github.com/Satyajit-Mohajan/E-Commerce-Sales-Data-Analysis-Using-Hadoop>

7. REFERENCE

<https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop>